

Utkarsh Arora

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TECHNICAL SKILLS

Programming Languages: Python (Expert), SQL, C++, C#, JavaScript (ES6+), Java

AI/ML Frameworks & Libraries: TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn, MediaPipe, Gemini API

Web Technologies & Frameworks: React, Node.js, Express.js, ASP.NET Core, Spring Boot, Tailwind CSS, RESTful APIs, JWT, Stripe API

Databases & Cloud: PostgreSQL, SQL Server, MongoDB, Entity Framework Core, Cloud Basics (AWS/Azure)

Developer Tools & Methodologies: Git, GitHub, VS Code, CI/CD, Agile Methodologies, OOP, DSA, Statistical Analysis

EDUCATION

Bachelor of Technology

July 2023 - June 2027

Panipat Institute of Engineering and Technology, Panipat

- Specialization in Artificial Intelligence and Machine Learning

PROJECTS

Smart Traffic Management System

[Github Link](#)

Smart Traffic Management System using Reinforcement Learning

- **Developed** a Reinforcement Learning (Deep Q-Learning) platform to optimize urban traffic flow.
- **Implemented** a custom reward function that dynamically adjusts signal timings based on real-time vehicle density, achieving a **25% reduction in traffic simulation delay**.
- **Integrated** a real-time data visualization dashboard to monitor agent performance and traffic metrics.
- **Designed** a modular architecture allowing for easy integration of different intersection geometries.

Technologies Used: Python, TensorFlow/Keras, SUMO, TraCI, Matplotlib, NumPy, Node.js, React, Express, MongoDB, Tailwind CSS

Diabetes Predictor

[Github Link](#)

A machine learning-based diagnostic tool designed to predict the likelihood of diabetes in patients based on clinical health metrics.

- **Built** a machine learning (Support Vector Machines) classification model using the Pima Indian Diabetes Dataset.
- **Achieved** a prediction **accuracy of 87%** after comprehensive data preprocessing and hyperparameter tuning.
- **Integrated** a user-friendly interface for instantaneous risk assessments, ready for **Deployment** to aid healthcare providers.

Technologies Used: Python, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn.

Roshni Creations E-Commerce Platform

[Github Link](#)

- **Developed** a premium, full-stack e-commerce application (Group Project).
- Integrated a Gemini-powered conversational chatbot for intelligent customer support and led the development of a WebAR virtual try-on feature.
- Integrated a Gemini-powered conversational chatbot for intelligent customer support.

Technologies Used: React, Vite, Tailwind CSS, Java Spring Boot, PostgreSQL, Spring Security, JWT, Stripe API, Gemini API, MediaPipe.

EXPERIENCE

Intern | Power Grid Corporation of India Limited

June 2026 – July 2026

- Engineered the I2P Portal, a full-stack role-based application for PowerGrid's HRD, streamlining maintenance order workflows and field data management.
- Architected back-end services using ASP.NET Core and EF Core, ensuring high availability and 100% data integrity through robust RBAC security protocols.
- Designed responsive, data-driven dashboards with Chart.js and Tailwind CSS, increasing field data entry efficiency by 40% through optimized UI/UX.

Data Science and Machine Learning Intern | The BrainHub

June 2025 - August 2025

- Leveraged Python and machine learning libraries such as TensorFlow/Keras, Scikit-Learn, Pandas, and NumPy to build predictive models and analyze complex datasets.
- Applied advanced feature engineering and machine learning techniques to optimize model performance.

Data Analytics Intern | Grades Global

June 2024 - August 2024

- Automated reporting workflows using Python and PowerBI, transforming raw data into actionable insights for executive decision-making.
- Developed interactive data visualization dashboards and automated reporting workflows to communicate trends effectively.